

Ulrich Valdes Tchomba Ngounou

Biostatistics, Health Data Modelling & Patient Flow Analysis

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Professional Summary

Dual Master's graduate in **Mathematical Epidemiology and Data Modelling** (AIMS-Cameroon) and **Mathematical Modeling and Scientific Computing** (ENSP, University of Yaoundé I). Strong expertise in **biostatistics**, **stochastic processes** (Poisson processes, continuous-time Markov chains, queueing theory), **Bayesian inference**, and **statistical modelling of complex health systems**. Experienced in developing and implementing novel statistical methods, parameter estimation, and numerical simulations using R, Python, and MATLAB. Currently a research intern at the Higher Institute of Scientific and Medical Research (ISM), applying statistical and machine learning methods to real-world health data. Highly motivated to pursue a PhD in Statistics, with a strong interest in **modelling patient flow**, **hospital systems**, and **impact evaluation** to improve healthcare efficiency and patient outcomes.

Research Experience

Research Intern – Statistical Modelling of Neglected Tropical Diseases 2026 – Present
Higher Institute of Scientific and Medical Research (ISM) – Cameroon

- Adapting the SIMPOP model to evaluate the "slash and clear" strategy against onchocerciasis and loiasis using real field data.
- Implementing Approximate Bayesian Computation (ABC) for parameter calibration and seasonal analysis in Python/R.
- Applying statistical and machine learning methods to complex, non-stationary real-world health data.

Master's Thesis 2 – Stochastic Epidemiological Modelling 2025 – 2026
AIMS-Cameroon – Prof. J. C. Loredó-Osti, Memorial University, Canada (Supervisor)

- Developed a stochastic SIRWS model with waning immunity and periodic transmission.
- Used continuous-time Markov chains (CTMC) and Monte Carlo simulations to capture randomness in dynamic systems – **directly transferable to modelling patient transitions and hospital flow**.
- Implemented in R (rstan) and Python.

Master's Thesis 1 – Stability Analysis of Nonlinear PDE Systems 2023 – 2025
ENSP, University of Yaoundé I – Dr. Ivic Valaire Yatat (Supervisor)

- Spectral stability analysis of a coupled nonlinear PDE system using Fourier transforms and Taylor expansions.
- Developed rigorous mathematical proofs for stability thresholds.
- Advanced numerical simulations in MATLAB – directly applicable to complex dynamical systems.

Capstone Project – Medical Cost Prediction (Industry Immersion Program) 2026
iiAfrica (Cape Town, South Africa)

- Built end-to-end regression pipelines (Linear Regression, Random Forest, XGBoost) on 1,338 records.
- Performed advanced EDA and hypothesis testing; uncovered a strong interaction that raised model R^2 from 0.75 to 0.89.

AI Hackathon – Intelligent Tutoring Platform February 2026
AIMS-Cameroon (African Institute for Mathematical Science)

- Designed and deployed an AI-powered tutoring prototype in 48 hours using Python, NLP, and neural networks.

Education

Master's Degree in Mathematical Epidemiology and Data Modeling 2025 – 2026
African Institute for Mathematical Sciences (AIMS) – Cameroon Thesis: Stochastic

Modelling of the Periodicity of Infectious Diseases – Extension of the SIR Model to the SIRWS Model with Waning Immunity and Periodic Transmission Rate

Supervisor: Prof. J. C. Loredó-Osti, Memorial University, Canada

Relevant Courses:

- Bayesian Statistics and Inference (MCMC, Hierarchical Models) – Prof. Z. Taib, Atlas Biostat, Sweden
- Stochastic Modelling of Infectious Diseases (CTMC, Monte Carlo) – Prof. J. C. Loredó-Osti, Memorial University, Canada
- Advanced Data Analysis Methods I & II (Regression, Multivariate Analysis, ANOVA, High-Dimensional Statistics) – Prof. P. Green, University of Bristol, UK; Prof. T. Chekouo, University of Minnesota, Canada
- Supervised and Unsupervised Machine Learning (Neural Networks, Classification) – Dr. Haissa, University of Regina, Canada
- Advanced Scientific Programming (Python, R, SageMath)

Master's Degree in Mathematical Modelling and Scientific Computing

2023 – 2025

National Advanced School of Engineering (ENSP), University of Yaoundé I – Cameroon

Thesis: Existence of Stationary Patterns in a Predator-Prey Model with Nonlocal Resource Consumption and Nonlocal Predation

Supervisor: Dr. Ivic Valaire Yatat, ENSPY–University of Yaoundé I

Relevant Courses:

- Parameter Identification Methods, Inverse Problems and Model Calibration
- Numerical Methods for Large Systems and Eigenvalue Problems
- Analysis of Dynamical Systems and Applications
- **Probability and Stochastic Processes – Poisson processes, queueing theory, and continuous-time Markov chains, directly relevant to hospital patient flow modelling**
- Mathematical Modeling and Scientific Computing (Python, MATLAB)

Bachelor's Degree in Applied Mathematics

2018 – 2021

University of Dschang – Cameroon

Conferences & Scientific Meetings

M2BE Conference – Mathematical Modelling for Biosciences and Engineering

Jul. 2025

International Conference – ENSP Yaoundé, Cameroon

SOCASAT – 5th Edition of the Cameroon Statistical Society

Jul. 2026

National Conference – Hôtel des Députés, Yaoundé, Cameroon

Certifications & Professional Development

Industry Immersion Program (IIP) – iiAfrica

2026

Cape Town, South Africa

- Business analytics, machine learning, design thinking, storytelling with data.
- Capstone project on real-world health data.

Office Secretarial Skills (Word, PowerPoint, Excel)

2022

Skills

Biostatistics & Health Data Modelling

Statistical modelling of health systems, patient flow analysis, impact evaluation, survival analysis foundations, epidemiological modelling, Bayesian inference, parameter estimation, model calibration and validation.

Stochastic Processes & Probability

Poisson processes, queueing theory, continuous-time Markov chains (CTMC), Monte Carlo simulation, probability theory, rigorous proofs, stability analysis, high-dimensional statistics.

Time Series & Functional Data Analysis

Foundations in functional analysis, dynamical systems, ODE/PDE systems, spectral analysis, Fourier transforms, stochastic processes.

Data Science & Machine Learning

Supervised and unsupervised learning, neural networks, regression, multivariate analysis, predictive modelling.

Scientific Computing & Programming

R (rstan, lme4, deSolve, adaptiveta, ggplot2), Python (NumPy, SciPy, Matplotlib, scikit-learn, pandas), MATLAB, SageMath, Git/GitHub, LaTeX, HPC foundations, C++ (foundations).

Languages

French (native), English (professional proficiency – full AIMS training in English).

Teaching Experience

Mathematics Teacher & Tutor

2021 – 2025

Private schools and exam preparation centers – Cameroon (in person), France (remote), Belgium (remote)

- Teaching mathematics and scientific computing.
- Competitive exam preparation (engineering, medical, teaching schools).
- Pedagogical coordination of an exam preparation center (2023–2025).
- Online and in-person tutoring.

References

Dr. YATAT Ivric Valaire – Master's M1 Supervisor

University of Yaoundé I – ENSPY

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Dr. BEKALO Daniel Biftu – AIMS-Cameroon

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Prof. Jean Jules Tewa – University of Yaoundé I

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